

Percentage

Percent: The term 'percent' is derived from the Latin word 'Per centum'. It implies "out of every hundred". The symbol '%' is used to denote percentage. For example, 15% means 15 out of 100. Each ratio can be expressed as a percentage.

For example, $\frac{1}{2}$ can be expressed as a percentage by multiplying by 100; $\frac{1}{2} \times 100 = 50\%$

A given percentage value can be converted to corresponding fraction by dividing by 100.

Example: $75\% = 75 \text{ out of hundred} = \frac{75}{100} = \frac{3}{4}$

Percentage fraction conversion chart:

$$\frac{1}{2} = 50\%$$

$$\frac{5}{6} = 83\frac{1}{3}\%$$

$$\frac{2}{9} = 22\frac{2}{9}\%$$

$$\frac{1}{3} = 33\frac{1}{3}\%$$

$$\frac{1}{7} = 14\frac{2}{7}\%$$

$$\frac{1}{10} = 10\%$$

$$\frac{2}{3} = 66\frac{2}{3}\%$$

$$\frac{2}{7} = 28\frac{4}{7}\%$$

$$\frac{1}{11} = 9\frac{1}{11}\%$$

$$\frac{1}{4} = 25\%$$

$$\frac{3}{7} = 42\frac{6}{7}\%$$

$$\frac{2}{11} = 18\frac{2}{11}\%$$

$$\frac{3}{4} = 75\%$$

$$\frac{1}{8} = 12\frac{1}{2}\%$$

$$\frac{1}{12} = 8\frac{1}{3}\%$$

$$\frac{1}{5} = 20\%$$

$$\frac{3}{8} = 37\frac{1}{2}\%$$

$$\frac{5}{12} = 41\frac{2}{3}\%$$

$$\frac{2}{5} = 40\%$$

$$\frac{5}{8} = 62\frac{1}{2}\%$$

$$\frac{1}{15} = 6\frac{2}{3}\%$$

$$\frac{3}{5} = 60\%$$

$$\frac{7}{8} = 87\frac{1}{2}\%$$

$$\frac{1}{20} = 5\%$$

$$\frac{4}{5} = 80\%$$

$$\frac{1}{9} = 11\frac{1}{9}\%$$

$$\frac{1}{25} = 4\%$$

$$\frac{1}{6} = 16\frac{2}{3}\%$$

Formula to calculate percentage value: $y\% \text{ of } x = \left(\frac{y}{100}\right) \times x$

From the above formula, we have the following result: $x\% \text{ of } y = y\% \text{ of } x$.

whenever we have two numbers x and y , one number can be expressed as a percentage of the other as follows:

x as a percent of $y = \frac{x}{y} \times 100$, y as a percent of $x = \frac{y}{x} \times 100$.

Percentage increase or decrease:

$$\text{Percentage increase} = \frac{\text{increase in the quantity}}{\text{original quantity}} \times 100$$

$$\text{Percentage decrease} = \frac{\text{decrease in the quantity}}{\text{original quantity}} \times 100$$

For example, if the net profit of a company grew from 50 crore in 2003 to 75 crore in 2004, then the percentage increase in the net profit from 2003 to 2004 is calculated as follows:

increase in the net profit = (75 – 50) crore = 25 crore

This increase is on Rs. 50 crore.

$$\text{So, Percentage increase} = \frac{\text{increase in profit from 2003 to 2004}}{\text{Net profit in 2003}} \times 100 = \frac{25}{50} \times 100 = 50\%$$

When a quantity increases or decreases by some percent, the new value of the quantity can be directly calculated as follows:

If the original quantity is 120 and it increases by 25% , then the new quantity is: $1.25 \times 120 = 150$

(Here, $1.25 = 1 + 0.25$, where 0.25 is equivalent to 25%)

Similarly, if there is a decrease by 25% on 120, then the new quantity is: $0.75 \times 120 = 90$

(Here, $0.75 = 1 - 0.25$, where 0.25 is equivalent to 25%)

Some important conclusions:

(i) If x is a% more than y, then y is $\left(\frac{a}{100 + a} \times 100 \right)\%$ less than x.

(ii) If x is a% less than y, then y is $\left(\frac{a}{100 - a} \times 100 \right)\%$ more than x.

Example: If in an examination, the marks secured by Prerna are 20% less than that of Vinita, then marks secured by Vinita are how much percent more than prerna's marks?

Solution: a = 20%

According to the above formula; Required percentage = $\left(\frac{a}{100 - a} \times 100 \right)\% = \frac{20}{80} \times 100 = 25\%$

(iii) If a number is first increased by a% and then decreased by a% then the net effect is always a decrease which is equal to 'a% of a' i.e., $\frac{a^2}{100}\%$

Example: The salary of a worker is first increased by 5% and then it is decreased by 5%. What is the change in his salary?

Solution: Here a = 5%

There will be a net decrease; Percent decrease = $\frac{a^2}{100}\% = \frac{5^2}{100}\% = 0.25\%$

(iv) If a quantity is first changed (increased or decreased) by a% and then changed (increased or decreased) by b%, then

$$\text{Net change} = \left[\pm a \pm b + \frac{(\pm a)(\pm b)}{100} \right]\%$$

Net change is an increase or a decrease according to the positive or negative sign, respectively of the final result.

Example: The price of an article is first increased by 20% and then decreased by 25% due to reduction in sales. Find the net percent change in the final price of the article.

Solution: a = 20%, b = 25%

$$\text{Required percentage change} = \left(20 - 25 + \frac{20 \times (-25)}{100} \right)\% = (-5 - 5)\% = -10\%$$

So, there is a net decrease of 10% in the final price of the article as the final result is negative.

(v) If the price of a commodity increases or decreases by a%, then the decrease or increase in consumption, so as

not to increase or decrease the expenditure is equal to $\left(\frac{a}{100 \pm a}\right) \times 100\%$

(vi) If the population of a town is P and it increases (or decreases) at the rate of R% per annum, then

(i) Population after n years = $P \left(1 \pm \frac{R}{100}\right)^n$

(ii) Population n years ago = $\frac{P}{\left(1 \pm \frac{R}{100}\right)^n}$

('+' sign for increment; '-' sign for decrement).

Some tricks to calculate faster:

(i) Splitting the percentage into parts

Example: Find 51% of 128.

Solution: 51% of 128 = (50 + 1)% of 128 = 50% of 128 + 1% of 128 = 64 + 1.28 = 65.28

(ii) Interchanging the percentage value and the number

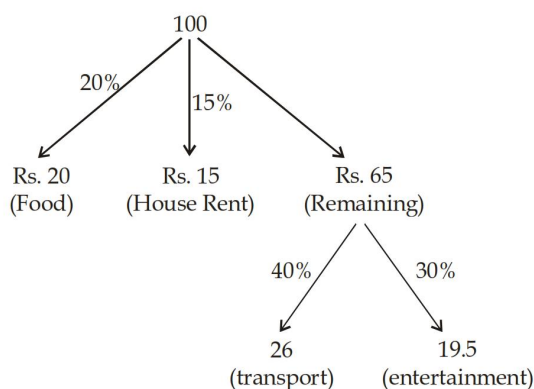
Example: Find 39% of 12.5

Solution: 39% of 12.5 = 12.5% of 39 = $\frac{1}{8} \times 39 = 4.875$

Types of Questions

1. Nikhil spent 20% of his monthly income on food and 15% on house rent. 40% of the remaining he spent on transport and 30% on entertainment. He is left with an amount of Rs. 8775 after all the expenditures. What is Nikhil's monthly income?

Sol. Let the income be Rs. 100



Final Saving = 100 - 20 - 15 - 26 - 19.5 = Rs. 19.5; But his final saving is Rs. 8775

$$19.5 \xrightarrow{(\times 450)} 8775$$

So, Income = Rs. 100 $\times$ 450 = Rs. 45000.

2. If the price of a Commodity be raised by 40%, by how much percent must a householder reduce his consumption of that commodity, so as not to increase his expenditure?

Sol. Here, a = 40%; According to the formula, Reduction in Consumption

$$= \left(\frac{40}{100 + 40} \times 100 \right) \% = \frac{40}{140} \times 100$$

$$= \frac{200}{7} \% = 28\frac{4}{7} \%$$

3. The population of a town is 352800. If it increases at the rate of 5% per annum, then what will be its population 2 years hence. Also, find the population 2 years ago.

Sol. P = 352800, R = 5%, n = 2

Population after 2 years

$$= P \left(1 + \frac{R}{100}\right)^n = 352800 \times \left(1 + \frac{5}{100}\right)^2$$

$$= 352800 \times \frac{21}{20} \times \frac{21}{20} = 388962$$

Population 2 years ago

$$= \frac{P}{\left(1 + \frac{R}{100}\right)^n} = \frac{352800}{\left(1 + \frac{5}{100}\right)^2}$$

$$= 352800 \times \frac{20}{21} \times \frac{20}{21} = 320000$$

- $$32 \xrightarrow{(\times 200)} 6400 \Rightarrow 6400 \text{ votes}$$

- $$\text{Required percentage} = \frac{60}{75} \times 100 = 60 \times \frac{4}{3} = 80\%$$

- (a) 35% (b) 65%
(c) 285.71% (d) 420%

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16. If 35% of a number is 12 less than 50% of that number, then the number is:
 (a) 40 (b) 50
 (c) 60 (d) 80
17. What percentage of numbers from 1 to 70 have squares that end in the digit 1?
 (a) 1% (b) 14%
 (c) 20% (d) 21%
18. If 75% of a number is added to 75, then the result is the number itself. The number is:
 (a) 50 (b) 60
 (c) 300 (d) 400
19. The sum of two numbers is 2490. If 6.5% of one number is equal to 8.5% of the other, then the numbers are:
 (a) 989, 1501 (b) 1011, 1479
 (c) 1401, 1089 (d) 1411, 1079
20. When any number is divided by 12, then dividend comes $\frac{1}{4}$ th of the other number. By how much percent first number is greater than the second number?
 (a) 150 (b) 200
 (c) 300 (d) Data inadequate
21. If one number is 80% of the other and the sum of their square is 656, then the numbers are:
 (a) 4, 5 (b) 8, 10
 (c) 16, 20 (d) 14, 12
22. A person's salary has increased from Rs. 7200 to Rs. 8100. What is the percentage increase in his salary?
 (a) 25% (b) 18%
 (c) $16\frac{2}{3}\%$ (d) $12\frac{1}{2}\%$
23. A's salary is 20% less than B's salary. Then B's salary is more than A's salary by:
 (a) $33\frac{1}{2}\%$ (b) $16\frac{2}{3}\%$
 (c) 20% (d) 25%
24. If the price of petrol is increased by 20%, by what percentage should the consumption be decreased by the consumer, if the expenditure on petrol remains unchanged?
 (a) $16\frac{2}{3}\%$ (b) $6\frac{2}{3}\%$
 (c) 8% (d) 15%
25. The price of an article of Rs. 100. Its price is increased by 10%, then again its price is increased by 10%. How much is increased in total price?
 (a) 20 (b) 21
 (c) 110 (d) 121
26. An agent sells goods of value of Rs. 15000. The commission which he receives is $12\frac{1}{2}\%$, is what amount?
 (a) Rs. 1875 (b) Rs. 2125
 (c) Rs. 2000 (d) Rs. 2700
27. The monthly income of a person is Rs. 5000. If his income is increased by 30%, then what is his monthly income now?
 (a) Rs. 7000 (b) Rs. 5500
 (c) Rs. 4500 (d) Rs. 6500
28. The price of a certain article is Rs. 15000. But due to slump in the market, its price decrease by 8%. Find the new price of the article?
 (a) Rs. 14000 (b) Rs. 13800
 (c) Rs. 16500 (d) Rs. 12600
29. A man losses 20% of his money. After spending 25% of the remainder, he has Rs. 480 left. What is the amount of money he originally had?
 (a) Rs. 600 (b) Rs. 720
 (c) Rs. 720 (d) Rs. 800
30. 15% of 10% of 20% of 1000 is?
 (a) 1.50 (b) 67
 (c) 150 (d) 3
31. If the numerator of a fraction is increased by 120% and denominator is also increased by 350% then the fraction become $\frac{11}{27}$ what was fraction?
 (a) $\frac{4}{5}$ (b) $\frac{5}{6}$
 (c) $\frac{6}{5}$ (d) $\frac{5}{4}$
32. If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number?
 (a) 2 : 5 (b) 3 : 7
 (c) 5 : 3 (d) 7 : 3
33. A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?
 (a) 45% (b) $45\frac{5}{11}\%$
 (c) $54\frac{6}{11}\%$ (d) 55%
34. If 50% of (x-y)=30% of (x+y), then what percent of x is y?
 (a) 20% (b) 25%
 (c) 30% (d) 40%
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35. In expressing a length 81.472 km as nearly as possible with three significant digits, find the percentage error?
 (a) 0.1% (b) 0.034%
 (c) 0.0020% (d) 0.4%
36. Difference of two numbers is 1600. If 7.5% of one number is 12.5% of the other number, find the two numbers?
 (a) 2490, 4000 (b) 2400, 4000
 (c) 2490, 4150 (d) 2500, 4200
37. Sixty-five percent of a number is 21 less than four-fifth of that number. What is the number?
 (a) 100 (b) 120
 (c) 140 (d) 160
38. An inspector rejects 0.08% of the meters as defective. How many will he examine to reject 2?
 (a) 2000 (b) 2200
 (c) 2500 (d) 2800
39. Which is greatest number?
 (a) $16\frac{2}{3}\%$ (b) $\frac{2}{15}$
 (c) $\frac{1}{11}$ (d) 0.17
40. $16\frac{2}{3}\%$ of 600 gm - $33\frac{1}{3}\%$ of 180 gm
 (a) 20 gm (b) 30 gm
 (c) 40 gm (d) 60 gm

Moderate

1. In an examination it is required to get 35% of the aggregate marks to pass. Rishu got 216 marks and declared failed by 5% marks then what was the total marks?
 (a) 620 (b) 720
 (c) 820 (d) 710
2. The price of Petrol went up 20%. In order to keep expenses same Ram must reduce travel by:
 (a) 25% (b) 30%
 (c) 33.33% (d) 16.67%
3. The length and breadth of a rectangle field are increased, the area increased by 50%. If the length increased by 20%, by what percentage was breadth increased?
 (a) 25% (b) 30%
 (c) 20% (d) Data inadequate
4. Aditya's salary is 80% of Amit's salary and 120% of Rajiv's salary. What is Amit's salary if Rajiv's salary is 30,000?
 (a) 40000 (b) 45000
 (c) 50000 (d) 55000
5. In a class 60% of the students pass in Hindi and 45% pass in Sanskrit. If 25% of them pass in both subjects, what percentage of the student fails in both subjects?
 (a) 80% (b) 20%
 (c) 25% (d) 75%
6. An arc contains 23% copper. To get 69% of copper the quantity of the arc required is:
 (a) 200 (b) 250
 (c) 300 (d) 400
7. Rice is now being sold at Rs. 30 per kg. During last month its rate was Rs. 26 per kg. Find by how much percentage must a family reduce its consumption to keep the expenditure fixed?
 (a) $12\frac{1}{2}\%$ (b) $13\frac{1}{3}\%$
 (c) 14% (d) 15%
8. Aditya's salary was increased by 40% and then decreased by 25%. Find out net effect in Aditya's salary.
 (a) 4% increase (b) 3% increase
 (c) 5% increase (d) 6% increase
9. In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is $\frac{2}{3}$ of the number of students of 8 years age which is 48. What is the total number of students in the school?
 (a) 72 (b) 80
 (c) 100 (d) 150
10. A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?
 (a) 34% (b) 44%
 (c) 54% (d) 64%
11. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was?
 (a) 2700 (b) 2900
 (c) 3000 (d) 3100
12. Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of the total votes did the winning candidate get?
 (a) 57% (b) 60%
 (c) 65% (d) 90%

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13. Rajeev buys good worth Rs. 6650. He gets a rebate of 6% on it. After getting the rebate, he pay sales tax @ 10%. Find the amount he will have to pay for the goods?
 (a) Rs. 6876.10 (b) Rs. 6999.20
 (c) Rs. 6654 (d) Rs. 7000
14. The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is?
 (a) 4.37% (b) 5%
 (c) 6% (d) 8.75%
15. In an examination, A candidate obtains 25% marks and and fails by 45 marks while another candidate obtains 46% marks and passed by 15% marks. What are the passing marks?
 (a) 600 (b) 675
 (c) 700 (d) 750
16. Aditya has some amount out of which 25% is stolen in bus, 10% he gave to his friend, 50% of remainder is spent in party and rest Rs. 26 gave to his mother. What was his initial amount?
 (a) 1230 (b) 80
 (c) 160 (d) 90
17. If A's salary is 20% higher than B then how much B salary is less than A's salary?
 (a) 20% (b) 25%
 (c) 16.67% (d) $33\frac{1}{3}\%$
18. If a number is increased by 20% and then reduced by 20%. After this operation the number?
 (a) does not change (b) increase by 1%
 (c) 4% increase (d) 4% decrease
19. If $\frac{3}{5}$ of a number is 23 more than 50% of the same number, then what will be 80% of the number?
 (a) 230 (b) 174
 (c) 23 (d) 184
20. Two numbers are respectively 20% and 30% less than third number. What is the second number as a percentage of the first?
 (a) 87.5% (b) 88%
 (c) 77.5% (d) 87%
21. The sum of two number is $\frac{23}{20}$ of the first number. The second number is what percent of the first?
 (a) 5% (b) 10%
 (c) 15% (d) 20%
22. In an examination it is required to multiply a number with 7 but by mistake a Student divide it by 7 then what is error percent?
 (a) 7% (b) 97.96%
 (c) 91% (d) 49%
23. 405 toffees were distributed equally among children in such a way that the number of toffees received by each child is 20% of total number of children. How many toffees did each child got?
 (a) 40 (b) 30
 (c) 42 (d) 9
24. Ram scored 30% marks and failed by 15 marks. Aditya score 40% marks and obtained 35 marks more than those required to pass. The pass percentage is?
 (a) 33% (b) 38%
 (c) 43% (d) 46%
25. The price of sugar is reduced by 2%. How many kg of sugar can now be bought for the money which was sufficient to buy 49 kg of sugar earlier?
 (a) 1 kg less (b) 1 kg more
 (c) 2 kg more (d) 2 kg less
26. Aditya's salary is 125% of Ram's salary. Sanjay's salary is 80% of Ram's salary. If the total of all the three salaries is Rs. 61000. What is Sanjay's salary?
 (a) 10000 (b) 12000
 (c) 15000 (d) 16000
27. If A is 150% of B then B is what percent of A+B?
 (a) 30% (b) 35%
 (c) 40% (d) 45%
28. If in the examination 20% failed in Hindi, 25% failed in English and 7% failed in both subject then the percentage of student who failed in atleast one of the subjects?
 (a) 50% (b) 45%
 (c) 55% (d) 38%
29. If 30% students failed in English, 45% failed in Math and 25% failed in both then the student who pass in both subject is?
 (a) 38% (b) 55%
 (c) 70% (d) 50%
30. Population of Delhi increases by 10% every year. If the current population of Delhi is 1,331,000 then what was its population 3 years ago?
 (a) 1000000 (b) 25000
 (c) 10000000 (d) 1543200
31. A, B, C shared Rs. 18500 so that A received 25% more than B and B received 20% more than C then what amount did A received?
 (a) 7000 (b) 5000
 (c) 7500 (d) 8000
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32. If the price of a shirt is increased by 25% and then decreased by 30% then what will be net effect?
 (a) 12% increase (b) 12% decrease
 (c) 12.5% increase (d) 12.5% decrease
33. Nutan got 456 marks in an exam. Aditya got 54% marks in same exam which is 24 less than Nutan. The minimum passing marks in exam is 35%. Then how much marks more than passing marks did nutan get?
 (a) 280 (b) 456
 (c) 180 (d) 176
34. 1200 boy and 650 girls appeared in examination. If 70% of boy and 40% of girls failed. Find the approximate percentage of passed students?
 (a) 30% (b) 41%
 (c) 50% (d) Can't determine
35. Aditya's monthly salary is 15% more than Sanjay's Monthly salary if Aditya's monthly salary is 17250 than what is Sanjay's annual salary?
 (a) 150000 (b) 170000
 (c) 180000 (d) 250000
36. The population of Delhi two years ago was 55000. It increased by 12% in the first year and decreased by 15% in the second year. What was the population of town at the end of 2 years?
- (a) 7084 (b) 70804
 (c) 52360 (d) Can't determine
37. If the length of rectangle is increased by 20% and breadth is decreased by 10% then what will be impact on area?
 (a) 5% (b) 10% increase
 (c) 20% decrease (d) 8% increase
38. If the length of rectangle is increased by 20% and breadth is decreased by 10% then what will be impact on perimeter?
 (a) Can't be determined (b) 20% decrease
 (c) 5% increase (d) 8% increase
39. Ram gets 20% marks more than Girish. Girish get 20% more than Sanjay. Sanjay gets 20% less than Aditya. If Ram got 576 marks and total marks were 800 then what marks did Aditya get?
 (a) 600 (b) 480
 (c) 500 (d) 600
40. The product of one third of a number and 150% of another number is what percent of product of the given numbers?
 (a) 30% (b) 35%
 (c) 39% (d) 50%

Difficult

1. Ram borrowed Rs. 725 from Shyam at the beginning of a year at interest. After 8 months, he again borrowed Rs. 362.50 at a rate of interest double that the former sum bears. At the end of the year, the sum of interest on both loans is Rs. 43.50. Find the first rate of interest per annum?
 (a) 4.5% (b) 4.75%
 (c) 6.25% (d) 7.2%
2. The cost price of goods with a bankrupt is Rs. 25500 and if the goods had realised in their full value, his creditors would have received 85 paise in the rupee. But $\frac{2}{5}$ of the goods were sold at 17% and the remainder at 22% below their cost price. How many paise in a rupee was received by the creditors?
 (a) 72 paise (b) 68 paise
 (c) 55 paise (d) 52 paise
3. A Shopkeeper undertakes to supply 2000 tables at Rs. 1725 each. He estimates that if 10% are defective which will be sold at 50%, then the profit will be 15% on his whole outlay. When the tables were supplied, 70% of the tables were found defective. What loss did the Shopkeeper incur?
 (a) Rs. 607500 (b) Rs. 557500
 (c) Rs. 550500 (d) Rs. 80680
4. Sweta invested Rs. 10,000 in a scheme exactly three years ago. The value of the investment increased by 10% during the first year, increased by 5% during the second year, and decreased by 10% during the third year. What is the value of the investment today?
 (a) Rs. 10,500 (b) Rs. 10,395
 (c) Rs. 10,342 (d) Rs. 10,230
5. In Mumbai, 60% of the registered voters are BJP-supporters and the rest are Congress-supporters. In a mayoral race, if 75% of the registered voters who are BJP-supporters and 20% of the registered voters who are Congress-supporters are expected to vote for candidate X, what percent of the registered voters are expected to vote for candidate X?
 (a) 53% (b) 55%
 (c) 57% (d) 59%
6. A pharmaceutical company received Rs. 3 million in royalties on the first Rs. 20 million in sales of the generic equivalent of one of its products and then Rs. 9 million in royalties on the next Rs. 108 million in sales. By approximately what percent did the ratio of royalties to sales decrease from the first Rs. 20 million in sales to the next Rs. 108 million in sales?
 (a) 10.27% (b) 20.63%
 (c) 38.6% (d) 44.44%

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7. In Jamshedpur, only two newspapers Dainik Jagran and Prabhat Khabar are published. It is known that 25% of the city population reads Dainik Jagran and 20% reads Prabhat Khabar while 8% reads both the newspapers. It is also known that 30% of those who read Dainik Jagran but not Prabhat Khabar look into advertisement and 40% of those who read Prabhat Khabar but not Dainik Jagran look into advertisement while 50% of those who read both the newspapers look into advertisements. What is the percentage of the population who read an advertisement?
- (a) 13.9% (b) 15.8%
(c) 17.2% (d) 21.4%
8. In my office, at least 50% of the people read an e-newspaper. Among those who read an e-newspaper, at most 25% read more than one e-paper. Only one of the following statements follows from the statements given below. Which one is it?
- (a) At the most 37.5% read exactly one e-paper.
(b) At least 37.5% read exactly one e-paper.
(c) At the most 19.8% read exactly one e-paper.
(d) At least 19.8% read exactly one e-paper.
9. In Convent Model School, 60% of the students are boys. In an aptitude test, 80% of the girls scored more than 40 marks (out of a maximum possible 150 marks). If 60% of the total students scored more than 40 marks in the same test, find the fraction of the boys who scored 40 marks or less?
- (a) $\frac{3}{5}$ (b) $\frac{6}{7}$
(c) $\frac{5}{7}$ (d) $\frac{7}{15}$
10. In a recent opinion poll held during April, 60% of the respondents favoured India Against Corruption (IAC) while the rest favoured Indian political parties (IPP). It was found in May polls that 10% of IAC supporters switched their preference to IPP, while the same percentage of IPP's supporters also switched their preference to IAC. What percentage of the electorate should now switch their preference from IAC to IPP so that they are at par?
- (a) 14% (b) 19%
(c) 24% (d) 29%
11. Suman's project report on 'Development with dignity', consists of 25 pages each of 60 lines with 75 characters on each line. In case the number of lines is reduced to 55 but the number of characters is increased to 90 per lines, what is the percentage change in the number of pages. (Assume the number of pages to be a whole number.)
- (a) - 8% (b) + 8%
(c) + 12% (d) 80%
12. Visions Pvt. Ltd. Appoints a sales representative on the basic salary of Rs. 1200 per month and the condition that for every sales of Rs. 10000 above Rs.10000, he will get 50% of basic salary and 10% of the sales as a reward. There is no incentive for the first Rs. 10000 of sales. What should be the value of sales if the sales representative wants to earn Rs. 7600 in a particular month?
- (a) Rs. 120000 (b) Rs. 50000
(c) Rs. 80000 (d) Rs. 45000
13. Neha has a watch which gain 2% per hour when the temperature is in the range of 40°C–50°C and it loses at the same rate when the temperature is in the range of 20°C – 30°C. The watch runs on times in all other temperature ranges. On a sunny day, the temperature started soaring up from 8 a.m. in the morning at the uniform rate of 2°C per hour and during the afternoon it started coming down at the same rate. Find what time will it be by the watch at 7 p.m. if at 8 a.m. the temperature was 32°C and at 4 p.m. it was 40°C?
- (a) 5 : 12 : 42 p.m. (b) 6 : 28 : 33 p.m.
(c) 7 : 04 : 48 p.m. (d) None of these
14. In laptop market, only three competitors (Lenovo, Apple and Samsung) exist. Last year the sales of Apple laptops were 10% more than Lenovo. In year, both the firms Lenovo and Apple increased their respective sales by 20%. This year, the sales of the firm Apple are five times that of Samsung. How much were the sales of the firm (approx.) Samsung last year, if the total sales remained constant over the two years period?
- (a) 25% (b) 32%
(c) 38% (d) 41%
15. Two jars contain equal quantities of 40% alcohol. Swati changed the concentration of the first jar to 50% by adding extra quantity of pure alcohol. Sonali changed the concentration of the second jar to 50% replacing a certain quantity of the solution with pure alcohol. By what percentage is the quantity of alcohol added by Swati more than that replaced by Sonali?
- (a) 10% (b) 20%
(c) 30% (d) 40%
16. For admission in a post graduate program of Calcutta University, 90% of the candidates who appeared for the written test were males and the rest were females, 60% of the males and 80% of the females passed in the written test. What is the total number of students who appeared for the written test, if the total number of passed candidates was 1240?
- (a) 1380 (b) 1560
(c) 2000 (d) 2500
-

17. In a gram panchayat meeting, 1000 people voted on a resolution with 10% of the votes being invalid. After some discussion 1000 people voted again. This time there were 20% invalid votes. The opponents were increased by 50% while the motion was now rejected by a majority, which is 300% more than it was formerly passed by. How many people voted against the resolution before the discussion?
 (a) 700 (b) 600
 (c) 500 (d) 400
18. An index of 12 shares contains, among others, the shares of Vision Power, Vision Infra and Vision Communication with weightage of 7%, 13% and 15% respectively. What is the increase in the prices of other shares, if these three rise by 9%, 10% and 4% respectively, while the index rises by 6%?
 (a) 5.34% (b) 5.94%
 (c) 6.23% (d) Can't be determine
19. A, B and C start a business by investing Rs. 70000 that earns them a profit of Rs. 42000 at the end of the

year. A invests his share in the profit in a scheme that gives her 10% interest compounded annually and B invests his share in a scheme that gives her 20% interest compounded annually. A gets Rs. 2520 as interest at the end of 2 years and B gets an interest of Rs. 4200 at the end of one year. Find C investment in the business?

- (a) Rs. 10000 (b) Rs. 15000
 (c) Rs. 20000 (d) Rs. 25000
20. Sashi has Rs. 90000 with him. He purchases a mobile, an i-pad and a laptop for Rs. 15000, Rs. 13000 and Rs. 35000 respectively and puts the remaining money in his bank account which pays 15% per annum compound interest. After 2 years he sells off the three items at 80% of their original price and also withdraws his entire money from the bank by closing the account. What is the total change in his asset?
 (a) 5.31% (b) 4.31%
 (c) 4.32% (d) - 4.32%

Previous Year (Memory Based)

1. In a test a student got 30% marks and failed by 25 marks. In the same test another student got 40% marks and secured 25% marks more than the essential minimum pass marks. The maximum pass marks for the test were :
 (a) 400 (b) 480
 (c) 500 (d) 580
2. In an examination, 80% of the boys passed in English and 85% passed in Mathematics, while 75% passed in both. If 45 boys failed in both. The number of boys who sat for the examination was:
 (a) 400 (b) 450
 (c) 200 (d) 150
3. A student scored 32% marks in science subjects out of 300. How much should he score in language papers out of 200 if he is to get overall 46% marks?
 (a) 72% (b) 67%
 (c) 66% (d) 60%
4. Ram spends 40% of his salary on food, 20% on house rent, 10% on entertainment and 10% on conveyance. If his savings at the end of a month are Rs 1500, then his salary per month (in Rs) is:
 (a) Rs 8000 (b) Rs 7500
 (c) Rs 6000 (d) Rs 10000
5. A man had a certain amount with him. He spent 20% of that to buy an article and 5% of the remaining on transport. Then he gifted Rs 120. If he is left with Rs 1400 the amount he spent on transport is :
 (a) Rs 76 (b) Rs 61
 (c) Rs 95 (d) Rs 80
6. 8% of the voters in an election did not cast their votes. In this election, there were only two candidates. The winner by obtaining 48% of the total votes, defeated his contestant by 1100 votes. The total number of voters in the election was?
 (a) 21000 (b) 23500
 (c) 22000 (d) 27500
7. A district has 64000 inhabitants. If the population increases at the rate of $2\frac{1}{2}$ % per annum, the number of inhabitants at the end of 3 years will be :
 (a) 70000 (b) 69200
 (c) 68921 (d) 68911
8. The value of a machine is Rs 6,250. It decreases by 10% during the first year, 20% during the second year and 30% during the third year. What will be the value of the machine after 3 years ?
 (a) Rs 2,650 (b) Rs 3,050
 (c) Rs 3,150 (d) Rs 3,510
9. 75 gm of sugar solution has 30% sugar in it. Then the quantity of sugar that should be added to the solution to make the quantity of the sugar 70% in the solution is :
 (a) 125 gm (b) 100 gm
 (c) 120 gm (d) 130 gm
10. The expenses of rice, fish and oil of a family are in the ratio 12 : 17 : 3. The price of these articles are increased by 20%, 30% and 50% respectively. The total expenses of family on these articles are increased by :

-
- (a) $14\frac{1}{8}\%$ (b) $7\frac{1}{8}\%$ (c) $9\frac{1}{11}\%$ (d) $11\frac{1}{11}\%$
- (c) $28\frac{1}{8}\%$ (d) None of these
11. A spider climbed $62\frac{1}{2}\%$ of the height of the pole in one hour and in the next hour it covered $12\frac{1}{2}\%$ of the remaining height. If pole's height is 192 m, then the distance climbed in second hour is:
- (a) 3 m (b) 5 m
(c) 7 m (d) 9 m
12. A and B are two fixed points 5 cm apart and C is a point on AB such that AC is 3 cm. If the length of AC is increased by 6%, the length of CB is decreased by:
- (a) 6% (b) 7%
(c) 8% (d) 9%
13. In an examination in which full marks were 500. A got 25% more than C, C got 20% less than D. If A got 360 marks. What percentage of full marks was obtained by D?
- (a) 72% (b) 80%
(c) 50% (d) 60%
14. The population of a town is 3,11,250. The ratio of women to men is 43:40. If there are 24% literate among women and 10% illiterate, among men, the total number of literate persons in the town is:
- (a) 1,70,700 (b) 1,73,700
(c) 1,75,700 (d) 1,73,200
15. The population of a village is 25,000. One-fifth are females and the rest are males, 5% of males and 40% of females are uneducated. What percentage on the whole are educated?
- (a) 75% (b) 88%
(c) 55% (d) 85%
16. In two successive years, 80 and 60 students of a school appeared at the final examination, of which 60% and 80% passed respectively. The average rate of students passed (in percent) is:
- (a) 68% (b) $68\frac{4}{7}\%$
(c) 32% (d) 36%
17. If the income of Ram is $12\frac{1}{2}\%$ more than that of Shyam, the income of Shyam is less than that of Ram by:
- (a) $11\frac{1}{9}\%$ (b) $12\frac{1}{8}\%$
18. If 30% of A = 0.25 of B = $\frac{1}{5}$ of C, then A : B : C is equal to:
- (a) 5 : 6 : 4 (b) 5 : 24 : 5
(c) 6 : 5 : 4 (d) 10 : 12 : 15
19. If 40% of (A + B) = 60% of (A - B) then $\frac{2A - 3B}{A + B}$ is.
- (a) $\frac{7}{6}$ (b) $\frac{6}{7}$
(c) $\frac{5}{6}$ (d) $\frac{6}{5}$
20. 25% of 120 + 40% of 380 = ? of 637:
- (a) $\frac{2}{7}$ (b) $\frac{1}{7}$
(c) $\frac{4}{7}$ (d) $\frac{3}{7}$
21. If 20% of A = 30% of B = $\frac{1}{6}$ of C, then A : B : C is:
- (a) 2 : 3 : 16 (b) 3 : 2 : 16
(c) 10 : 15 : 18 (d) 15 : 10 : 18
22. A person who spends $66\frac{2}{3}\%$ of his income is able to save Rs 1200 per month. His monthly expenses (in Rs) is:
- (a) 1,200 (b) 2,400
(c) 3,000 (d) 3,200
23. The monthly income of a person was Rs 13,500 and his monthly expenditure was Rs 9,000. Next year his income increased by 14% and his expenditure increased by 7%. The percentage increase in his savings was:
- (a) 7% (b) 21%
(c) 28% (d) 35%
24. X has twice as much money as that of Y and Y has 50% more money than that of Z. If the average money of all of them is Rs 110, then the money, which X has is:
- (a) Rs 55 (b) Rs 60
(c) Rs 90 (d) Rs 180
25. Tulsiram's salary is 20% more than that of Kashyap. If Tulsiram saves Rs 720 which is 4% of his salary, then Kashyap's salary is.
- (a) Rs 15,000 (b) Rs 12,000
(c) Rs 10,000 (d) Rs 22,000
-

26. One-third of a number is 96. What will 67% of that number be ?
 (a) 192.96 (b) 181.44
 (c) 169.92 (d) 204.88
27. A number is divided into two parts in such a way that 80% of 1st part is 3 more than 60% of 2nd part and 80% of 2nd part is 6 more than 90% of the 1st part. Then the number is:
 (a) 125 (b) 130
 (c) 135 (d) 145
28. The price of an article is decreased by 10%. To restore its former value the new price must be increased by :
 (a) 10% (b) 11%
 (c) $11\frac{1}{8}\%$ (d) $11\frac{1}{9}\%$
29. A number is increased by $x\%$; to get back to the original number, it is to be reduced by?
 (a) $\frac{10x}{100+x}\%$ (b) $\frac{100}{10+x}\%$
 (c) $x\%$ (d) $\frac{x}{100+x}\%$
30. The price of an article was increased by $r\%$. Later the new price was decreased by $r\%$. If the latest price was Rs 1, then the original price was :
 (a) Rs 1 (b) $\text{Rs } \frac{1-r^2}{100}$
 (c) $\text{Rs } \frac{\sqrt{1-r^2}}{100}$ (d) $\text{Rs } \left(\frac{10000}{10000-r^2} \right)$
31. The Government reduced the price of sugar by 10 per cent. By this a consumer can buy 6.2 kg more sugar for Rs 837. The reduced price per kg of sugar is:
 (a) Rs 12.50 (b) Rs 13.00
 (c) Rs 13.50 (d) Rs 14.00
32. A reduction of 20% in the price of wheat enables Bhuvnesh to buy 5 kg more wheat for Rs 320. The original rate (in rupees per kg) of wheat was:
 (a) 16 (b) 18
 (c) 20 (d) 21
33. When the price of cloth was reduced by 25%, the quantity of cloth sold increased by 20%. What was the effect on gross receipt of the shop?
 (a) 5% increase (b) 5% decrease
 (c) 10% increase (d) 10% decrease
34. In a school 40% of the students play football and 50% play cricket. If 18% of the students neither play football nor cricket, the percentage of the students playing both is :
 (a) 40% (b) 32%
 (c) 22% (d) 8%
35. The percentage of metals in a mine of lead ore is 60%.
 Now the percentage of silver is $\frac{3}{4}$ of metals and the rest is lead. If the mass of ore extracted from this mine is 8000 kg, the mass (in kg.) of lead is:
 (a) 4763 (b) 4764
 (c) 4762 (d) 4761
36. If the numerator of a fraction is increased by 20% and the denominator is decreased by 5%, the value of the new fraction becomes $\frac{5}{2}$. The original fraction is:
 (a) $\frac{24}{19}$ (b) $\frac{3}{18}$
 (c) $\frac{95}{48}$ (d) $\frac{48}{95}$
37. A man invests a part of Rs 10,000 at 5% and the remainder at 6%. The 5% investment yields annually Rs 76.50 more than the 6% investment. The amount invested at 6% is:
 (a) Rs 3,600 (b) Rs 3,550
 (c) Rs 3,850 (d) Rs 4,000
38. The allowance of an employee constitutes 165% of his basic pay. If he receives Rs 11925 as gross salary, then his basic pay is (in Rs):
 (a) 4000 (b) 5000
 (c) 4500 (d) 5500
39. The price of sugar is increased by 20%. If the expenditure on sugar has to be kept the same as earlier, the ratio between the reduction in consumption and the original consumption is:
 (a) 1 : 3 (b) 1 : 4
 (c) 1 : 6 (d) 1 : 5
40. If the price of a commodity is increased by 50%. By what fraction must its consumption be reduced so as to keep the same expenditure on its consumption:
 (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{2}$ (d) $\frac{2}{3}$

Foundation

Solution

1. (d); 3.5 can be expressed as $(3.5 \times 100)\% = 350\%$

2. (d); $\frac{15}{100} \times 34 = 5.10$ Rs.

3. (b); $\frac{88}{100} \times 370 + \frac{24}{100} \times 210 - x = 118$
 $x = 325.6 + 50.4 - 118 = 376 - 118 = 258$

4. (c); $\frac{860}{100} \times 50 + \frac{50}{100} \times 860 = \frac{860}{100} (50 + 50) = 860$

5. (b); $\frac{60}{100} \times 264 = 158.40 \Rightarrow \frac{15}{100} \times 1056 = 158.40$

6. (d); Passing percentage = $\frac{252}{270} \times 100 = 93\frac{1}{3}\%$

7. (b); Required percentage = $\frac{1987.50}{2650} \times 100 = 75\%$

8. (a); Percentage = $\frac{3}{24} \times 100 = 12\frac{1}{2}\%$

9. (c); Litres of pure acid = $\frac{20}{100} \times 8 = 1.6$ litres

10. (b); Best percentage = $\frac{425}{500}$

11. (b); $\frac{15}{100 \times 100} \times \frac{100}{300} \times 10000 = \text{Rs. } 5$

12. (c); $\frac{45}{100} \times 1500 + \frac{35}{100} \times 1700 = \frac{x}{100} \times 3175$
 $\Rightarrow x = \frac{1270 \times 100}{3175} = 40$

13. (b); Let the worth of cloth be x

$$x \times \frac{2.5}{100} = 12.50 \Rightarrow x = \frac{12500}{25} \Rightarrow x = 500 \text{ Rs.}$$

14. (b); Worth of the house = $2800 \times \frac{7}{2} = 9800$ Rs.

15. (c); 35% is = 175 $\Rightarrow 100\% = \frac{175}{35} \times 100 = 500$

$$\text{Required percentage} = \frac{500}{175} \times 100 = 285.71\%$$

16. (d); Let the number is x

$$\frac{50}{100}x - \frac{35}{100}x = 12 \Rightarrow \frac{15x}{100} = 12$$

$$x = \frac{12 \times 100}{15} \Rightarrow x = 80$$

17. (c); Total such numbers = 14

$$\text{Required percentage} = \frac{14}{70} \times 100 = 20\%$$

18. (c); Let the number be x.

$$\frac{75}{100}x + 75 = x \Rightarrow x - \frac{3}{4}x = 75$$

$$\frac{x}{4} = 75 \Rightarrow x = 300$$

19. (d); Let the two numbers be x and y.

$$x + y = 2490 \Rightarrow \frac{65}{1000}x = \frac{85}{1000}y$$

$$x = \frac{17}{13}y \Rightarrow \frac{17}{13}y + y = 2490$$

$$\frac{30y}{13} = 2490 \Rightarrow y = \frac{2490 \times 13}{30}$$

$$y = 1079 \Rightarrow x = 2490 - 1079 \Rightarrow x = 1411$$

20. (b); Let the First and Second number be x and y

$$\frac{x}{12} = \frac{y}{4} \Rightarrow \frac{x}{y} = \frac{3}{1}$$

$$\text{Required percentage} = \frac{3-1}{1} \times 100 = 200\%$$

21. (c); Let the Numbers be x and y.

$$x \times \frac{80}{100} = y \Rightarrow \frac{4x}{5} = y$$

$$\frac{x}{y} = \frac{5}{4} \quad \left[\begin{array}{l} x = 5k \\ y = 4k \end{array} \right]$$

$$x^2 + y^2 = 656 \Rightarrow 25k^2 + 16k^2 = 656$$

$$k^2 = \frac{656}{41} \Rightarrow k = 4 \Rightarrow x = 4 \times 5 = 20$$

$$y = 4 \times 4 = 16$$

$$\text{Numbers are 16 and 20}$$

22. (d); Percentage increase in salary = $\frac{900}{7200} \times 100$

$$= \frac{100}{8} = 12\frac{1}{2}\%$$

23. (d); Let B's salary be = 100, A's salary be = 80

$$\text{Required percentage} = \frac{20}{80} \times 100 = 25\%$$

24. (a); Let the price of petrol be 100

Increase Price = 120

$$\text{Required percentage} = \frac{120-100}{120} \times 100$$

$$= \frac{100}{6} = 16\frac{2}{3}\%$$

25. (b); Increased Price = $\frac{110}{100} \times \frac{110}{100} \times 100 = 121$

$$\text{Required value} = 121 - 100 = 21$$

26. (a); Commission = $15000 \times \frac{25}{200} = \frac{15000}{8} = 1875$ Rs.

27. (d); Increase Monthly income = $5000 \times \frac{130}{100} = 6500$ Rs.

28. (b); New Price of the article = $15000 \times \frac{92}{100}$
= 13800 Rs.

29. (d); Money left = $100\% - 20\% - [100\% - 20\%] \times \frac{25}{100}$

$$= 100\% - 20\% - 20\% = 60\%$$

$$\therefore 60\% = 480 \Rightarrow 100\% = \frac{480}{60} \times 100 = \text{Rs. } 800$$

30. (d); $\frac{15}{100} \times \frac{10}{100} \times \frac{20}{100} \times 1000 = 3$

31. (b); Let the number will be $\frac{x}{y}$

$$\text{Increased number} = \frac{x + \frac{120}{100}x}{y + \frac{350}{100}y} = \frac{11}{27} \Rightarrow \frac{220x}{450y} = \frac{11}{27}$$

$$\Rightarrow \frac{x}{y} = \frac{11}{27} \times \frac{45}{22}$$

$$\text{Original Fraction} \Rightarrow \frac{x}{y} = \frac{5}{6}$$

32. (c); Let the Numbers be x and y.

$$\frac{40}{100}x = \frac{2}{3}y \Rightarrow \frac{2}{5}x = \frac{2}{3}y$$

$$\frac{x}{y} = \frac{5}{3} \Rightarrow x : y = 5 : 3$$

33. (b); Run scores from boundaries and sixes

$$= 3 \times 4 + 8 \times 6 = 60$$

Required percentage by Running between the

$$\text{wickets} = \frac{110-60}{110} \times 100 = \frac{500}{11} = 45\frac{5}{11}\%$$

34. (b); $\frac{50}{100}(x-y) = \frac{30}{100}(x+y)$

$$5x - 5y = 3x + 3y \Rightarrow 2x = 8y \Rightarrow \frac{y}{x} = \frac{1}{4}$$

$$\text{Required Percentage} = \frac{1}{4} \times 100 = 25\%$$

35. (b); Error = $(81.5 - 81.472) = 0.028$

$$\text{Required percentage} = \frac{0.028}{81.472} \times 100 = 0.034\%$$

36. (b); Let the number be x and y

$$x - y = 1600 \Rightarrow \frac{75}{1000}x = \frac{125}{1000}y$$

$$x = \frac{5}{3}y \Rightarrow \frac{5}{3}y - y = 1600$$

$$\frac{2y}{3} = 1600 \Rightarrow y = 2400$$

$$x = 1600 + 2400 = 4000$$

37. (c); Let the number be x

$$\frac{65}{100}x + 21 = \frac{4}{5}x \Rightarrow \frac{4x}{5} - \frac{13x}{20} = 21$$

$$\frac{16x - 13x}{20} = 21 \Rightarrow \frac{3x}{20} = 21 \Rightarrow x = 140$$

38. (c); It means that 0.08% of x = 2

$$\frac{8}{100 \times 100} \times x = 2 \Rightarrow x = \frac{2 \times 100 \times 100}{8}$$

$$\text{Required Number} = 2500$$

39. (d); Greatest Number = 0.17

40. (c); $16\frac{2}{3} \times \frac{1}{100} \times 600 - 33\frac{1}{3} \times \frac{1}{100} \times 180$

$$\Rightarrow \frac{1}{6} \times 600 - \frac{1}{3} \times 180 = 100 - 60 = 40 \text{ gm}$$

Moderate

1. (b); If Rishu failed by 5% marks which means he got 30% marks.

$$30\% \text{ marks} = 216 \Rightarrow 100\% \text{ marks} = \frac{216}{30} \times 100$$

$$\text{Total marks} = 720$$
2. (d); If the price of petrol has increase by 20%, it has gone up $\frac{1}{5}$ th of its earlier price.
 $\therefore$ The percentage of reduction in petrol that will maintain the amount of money spent on petrol

$$= \frac{1}{1+5} = \frac{1}{6}$$

$$= 16.67\% \Rightarrow \text{Reduction in travel} = 16.67\%$$
3. (a); Let the increased breath be x

$$20 + x + \frac{20x}{100} = 50 \quad [\% \text{ increase} = m + n + \frac{mn}{100}]$$

$$x + \frac{x}{5} = 30 \Rightarrow \frac{6x}{5} = 30$$

$$x = \frac{30 \times 5}{6} \Rightarrow x = 25\%$$
4. (b); Let amit's salary be x

$$\text{Aditya salary} = \frac{120}{100} \times 30,000 = 36,000 \text{ Rs.}$$

$$\frac{x \times 80}{100} = 36000 \Rightarrow x = \frac{36000 \times 100}{80} = 45,000 \text{ Rs.}$$
5. (b); The percentage of students fails in both subjects

$$= 100 - [(60 + 45) - 25] = 20\%$$
6. (c); Required quantity of arc = $69 \times \frac{100}{23} = 300$
7. (b); Percentage consumption Reduced = $\frac{4}{30} \times 100$

$$= \frac{40}{3} = 13\frac{1}{3}\%$$
8. (c); Let Aditya's salary was 100 Rs.

$$\text{Increased salary} = 140 \text{ Rs.}$$

$$\text{Decreased salary} = 140 \times \frac{75}{100} = 105 \text{ Rs.}$$

$$\text{Net effect} = 105 - 100 = 5\% \text{ increase}$$
9. (c); Student of 8 years = 48

$$\text{Total number of Students of age more than 8 years}$$

$$= 48 \times \frac{2}{3} = 32$$

$$\text{Total student 8 years old or above} = 32 + 48 = 80$$

$$\text{Now 20\% of students of is below 8 years old, so}$$

$$80\% \text{ of student} = 80$$

$$\text{So total student} = 100$$
10. (d); Let number is 15

$$\text{Original result} = 15 \times \frac{5}{3} = 25$$

$$\text{Wrong result} = 15 \times \frac{3}{5} = 9$$

$$\text{Percentage error} = \frac{25-9}{25} \times 100 = 64\%$$
11. (a); No of valid votes = $7500 \times \frac{80}{100} = 6000$

$$\text{No. of votes that other candidate got} = 6000 \times \frac{45}{100}$$

$$= 2700$$
12. (a); Total no of votes = $1136 + 7636 + 11628 = 20400$

$$\text{required percentage} = \frac{11628}{20400} \times 100 = 57\%$$
13. (a); Rebate = $6650 \times \frac{6}{100} = 399 \text{ Rs.}$

$$\text{Amount after Rebate} = 6650 - 399 = \text{Rs. } 6251$$

$$\text{Amount paid for the goods} = 6251 \times \frac{110}{100}$$

$$= 6876.10 \text{ Rs.}$$
14. (b); Percentage increase in decade = $\frac{87500}{175000} \times 100$

$$= 50\%$$

$$\therefore 50\% \text{ increase in 10 years.}$$

$$\therefore \text{Per years \% increase} = 5\%$$
15. (d); Passing percentage = $46\% - 15\% = 31\%$

$$\therefore 6\% = 45 \Rightarrow 100\% = \frac{45}{6} \times 100$$

$$\therefore \text{Total marks} = 750$$
16. (b); Let the initial amount be 100%

$$\text{Total amount stolen and gave to friend} = 35\%$$

$$\text{Remaining amount} = 65\%$$

$$\text{Amount spend on party} = \frac{65}{2} = 32.5\%$$

$$32.5\% = 26$$

$$100\% = \frac{26}{32.5} \times 100 \Rightarrow \text{Initial amount} = \text{Rs. } 80$$
17. (c); Let B's salary be 100, A's salary be 120

$$\text{Required pcentage} = \frac{20}{120} \times 100 = \frac{100}{6} = 16.67\%$$

18. (d); Let the number be 100, Increased number = 120
 Decreased number = $120 \times \frac{80}{100} = 96$
 After this number is decreased by 4%.
19. (d); Let the number be x
 $\therefore \frac{3}{5}x = 23 + \frac{50x}{100} \Rightarrow \frac{3}{5}x - \frac{1}{2}x = 23 \Rightarrow x = 230$
 $= \frac{80}{100} \times 230 = 184$
20. (a); Let the third number be 100
 Second number = 70, First number = 80
 Required percentage = $\frac{70}{80} \times 100 = 87.5\%$
21. (c); Let the two number be x and y
 $x + y = \frac{23}{20} \text{ of } x \Rightarrow y = \frac{23x}{20} - x \Rightarrow y = \frac{3x}{20}$
 Required percentage = $\frac{\frac{3x}{20}}{x} \times 100 = 15\%$
22. (b); Let the Number = 7
 So, correct answer = $7 \times 7 = 49$
 But answer obtained = $\frac{7}{7} = 1$
 $\% \text{ error} = \frac{48}{49} \times 100 = 97.96\%$
23. (d); Let the total number of children be x then
 $\frac{405}{x} = x \times \frac{20}{100} \Rightarrow \frac{405}{x} = \frac{x}{5} \Rightarrow x^2 = 2025$
 $x = 45$
 $\therefore \text{number of toffies each students get} = \frac{405}{45} = 9$
24. (a); 10% marks = $35 + 15 = 50$, 100 % marks = 50×10
 $\therefore \text{Total marks} = 500$
 Passing marks = $500 \times \frac{30}{100} + 15 = 165$
 Required percentage = $\frac{165}{500} \times 100 = 33\%$
25. (b); Let the price per kg is 100 rs.
 The total money used for 49 kg = 4900 Rs.
 After Price decrease = $\frac{4900}{98} = 50\text{kg}$
 Means 1 kg is the answer.
26. (d); Let Ram's salary is 100
 Sanjay's salary is 80 and Aditya's salary is 125
 Total of all salary = 305
 $\therefore \text{Sanjay's salary} = \frac{61000}{305} \times 80 = 16000 \text{ Rs.}$
27. (c); let B = 100, then A = 150
 Required percentage = $\frac{100}{250} \times 100 = 40\%$
28. (d); Required percentage = $20 + 25 - 7 = 38\%$
29. (d); percentage of students passed in both the subjects
 $= 100 - (30 + 45 - 25) = 50\%$
30. (a); Population x years ago = $\frac{A}{\left[1 + \frac{r}{100}\right]^n}$
 $= \frac{1331000}{\left[1 + \frac{10}{100}\right]^3} = \frac{1331000}{\frac{1331}{1000}} = 1000000$
31. (c); Let C received = 100
 $\therefore \text{B received} = 120$
 $\text{A received} = 120 \times \frac{125}{100} = 150$
 $\text{Amount A received} = \frac{18500}{370} \times 150 = \text{Rs. } 7500$
32. (d); Let the price of a shirt be 100
 25% increase = 125
 30 % decrease = $125 \times \frac{70}{100} = 87.5\%$
 $\therefore \% \text{ decrease} = 100 - 87.5 = 12.5\%$
33. (d); Marks of Aditya = $456 - 24 = 432$
 $\therefore 54\% = 432$
 $100\% = \frac{432}{54} \times 100 = 800$
 Passing marks = $800 \times \frac{35}{100} = 280$
 Required marks = $456 - 280 = 176$
34. (b); Total no of boy and girl appered in the examination = $1200 + 650 = 1850$
 No. of boy passed = $1200 \times \frac{30}{100} = 360$
 No. of girl passed = $650 \times \frac{60}{100} = 390$
 Total no. of boys and girl passed = 750

$$\text{Required \%} = \frac{750}{1850} \times 100 = 40.54\% \cong 41\%$$

35. (c); Sanjay's monthly salary = $\frac{17250}{115} \times 100$
 = Rs. 15000
 Sanjay's Annual salary = 15000×12
 = Rs. 180000

36. (c); Population at the end of 2 years.

$$55000 \times \frac{112}{100} \times \frac{85}{100} = 52360$$

37. (d); Impact on area

$$= 20 + (-10) + \frac{20 \times (-10)}{100} \left[\text{Using formula} \right]$$

$$m + n + \frac{mn}{100}$$

$$= 10 - 2 = 8\% \text{ increased}$$

38. (a); Can't be determined since impact on perimeter can't be determined as changes in different length and breadth give different net effect on perimeter.

39. (c); Let Sanjay gets 100 marks

∴ Girish gets 120 marks

$$\therefore \text{Ram gets} = 120 \times \frac{120}{100} = 144 \text{ marks}$$

$$\text{Sanjay marks} = \frac{576}{144} \times 100 = 400$$

$$\text{Aditya marks} = \frac{400 \times 100}{80} = 500 \text{ marks}$$

40. (d); Let the numbers be x and y

So product = xy

$$\text{given, } \frac{1}{3}x \times \frac{150y}{100} \Rightarrow \frac{1}{3}x \times \frac{3y}{2} \Rightarrow \frac{xy}{2}$$

$$\text{Required percentage} = \frac{\frac{xy}{2}}{xy} \times 100$$

$$= 50\%$$

Difficult

1. (a); $43.5 = \frac{725 \times R \times 1}{100} + \frac{362.5 \times 4 \times 2R}{12 \times 100}$

$$43.5 \times 300 = 2175R + 362.5 \times 2R = 2900R$$

$$R = 4.5$$

2. (b); Total debt = $22500 \times \frac{100}{85} = \text{Rs. } 30000$

Money received by selling the goods

$$= 25500 \left(\frac{2}{5} \times \frac{83}{100} + \frac{3}{5} \times \frac{78}{100} \right) = \frac{25500}{500} (166 + 234)$$

$$= 51 \times 400 = \text{Rs. } 20400$$

Therefore, money received by the creditors for a

$$\text{rupee} = \text{Rs. } \left(\frac{20400}{30000} \right) = \text{Rs. } 0.68 = 68 \text{ paise}$$

Hence, the creditor received 68 paise in a rupee.

3. (a); 10% of 2000 = 200

Selling price of 200 tables at 50%

$$= \text{Rs. } \left(200 \times \frac{1725}{2} \right) = \text{Rs. } 172500$$

Selling price of remaining 1800 tables

$$= \text{Rs. } (1800 \times 1725) = \text{Rs. } 3105000$$

Total revenue from selling 2000 tables

$$= \text{Rs. } (172500 + 3105000) = \text{Rs. } 3277500$$

Now, Rs. 3277500 includes 15% profit. Therefore,

$$\text{cost price of 2000 tables} = \frac{100}{115} \times 3277500$$

$$= \text{Rs. } 2850000$$

Now the actual selling price

$$= 2000 \times \frac{30}{100} \times 1725 + 2000 \times \frac{70}{100} \times \frac{1725}{2}$$

$$= 2000 \times 1725 \left(\frac{30}{100} + \frac{35}{100} \right)$$

$$= 20 \times 1725 \times 65 = \text{Rs. } 2242500$$

∴ Loss = Cost Price - Selling Price

Hence, the Shopkeeper incurs a loss of
 = Rs. 607500.

4. (b); The first year's increase of 10% can be expressed as 1.10; the second year's increase of 5% can be expressed as 1.05; and the third year's decrease of 10% can be expressed as 0.90. Now, multiply the original value of the investment account by each of these yearly changes.

$$10,000 \times 1.10 \times 1.05 \times 0.90 = 10395 \text{ Rs.}$$

Hence, the value of the investment today is
 = Rs. 10,395.

5. (a); Let y be the number of registered voters in Mumbai. Then, the information that 60% of the registered voters are from BJP can be expressed as $0.60y$. From this, it can be stated that $1.00y - 0.60y = 0.40y$ are from Congress. The percentage of BJP-supporters and the percentage of Congress-supporters who are expected to vote for candidate X can then be expressed as:
 $0.75 \times 0.60y + 0.20 \times 0.40y$
Simplify the expression to determine the total percentage of voters expected to vote for candidate X:
 $X = 0.75 \times 0.60y + 0.20 \times 0.40y$
 $= 0.45y + 0.08y = 0.53y$
Hence, 53% of the registered voters are expected to vote for candidate X.
6. (d); The ratio of royalties to sales for the first Rs. 20 million in sales is $\frac{3}{20}$, and the ratio of royalties to sales for the next Rs. 108 million in sales is $\frac{9}{108} = \frac{1}{12}$. The percent decrease in the royalties to sales ratios is 100 times the quotient of the difference in the ratios divided by the ratio of royalties to sales for the first Rs. 20 million in sales, i.e.,

$$\frac{\frac{1}{12} - \frac{3}{20}}{\frac{3}{20}} \times 100 = \left(\frac{1}{12} - \frac{3}{20} \right) \times \frac{20}{3} \times 100$$

$$= \left(\frac{5-9}{60} \right) \times \frac{20}{3} \times 100 = \frac{-4}{60} \times \frac{20}{3} \times 100 = \frac{-4}{9} \times 100$$

$$= -0.4444 \times 100 = -44.44 = 44.44\% \text{ decrease}$$
7. (a); Let the population of the city be 100. Then,
People reading Dainik Jagran = 25
People reading Prabhat Khabar = 20
People reading both = 8
People reading only Dainik Jagran = 17
People reading only Prabhat Khabar = 12
Therefore, required percentage of people who read an advertisement

$$= \left(17 \times \frac{30}{100} + \frac{12 \times 40}{100} + \frac{8 \times 50}{100} \right) = 13.9\%$$
8. (b); Let the number of people in my office = 100
At least 50 people read an e-newspaper.
At most 12.5 people read more than one e-newspaper.
Therefore, at least 37.5 people read only one e-newspaper.
Hence, at least 37.5% read exactly one e-newspaper.
9. (d); Let the total number of students be y . Then,
Number of boys = $\frac{3y}{5}$, Number of girls = $\frac{2y}{5}$
Number of girls scoring more than 40 marks
 $= \frac{4}{5} \times \frac{2y}{5} = \frac{8y}{25}$
Total number of students scoring more than 40 marks = $\frac{3y}{5}$
 $\therefore$ Required fraction = $\frac{7}{25} \times \frac{5}{3} = \frac{7}{15}$
Hence, the fraction of the boys who scored 40 marks or less is $\frac{7}{15}$.
10. (a); Let the total number of respondents = 100
People favoured IAC = 60
People favoured IPP = 40
New no. of people favouring IAC = 58
New no. of people favouring IPP = 42
Required percentage = 14%
11. (a); Total characters in the report = $25 \times 60 \times 75$
Let the new number of pages be n . Then,
 $n \times 55 \times 90 = 25 \times 60 \times 75$
 $\therefore n = \frac{25 \times 60 \times 75}{55 \times 90} = 22.72 \approx 23$
This means that Suman would need 23 pages for writing his report, which is a drop of 8% in terms of the pages.
12. (b); Income of the sales representative
 $= 1200 + 1600 \times y$
Where y is the number of Rs. 10000 sales he achieves over the initial Rs. 10000.
Therefore,
 $1200 + 1600 \times y = 7600$
Or, $y = \frac{(7600 - 1200)}{1600} \Rightarrow$ Or, $y = \frac{6400}{1600} = 4$
Thus, the total sales value must be Rs. 50000.
13. (c); At 12 noon, the watch would show the correct time, because till then the temperature range is below 40°C . The watch would gain 2% every hour between 12 noon and 4 p.m. An hour having 3600 seconds, it would gain 72 seconds in each of these hours. Thus, at 7 p.m. the watch would be $72 \times 4 = 288$ seconds ahead. The time exhibited by the watch would be 7 : 04 : 48 p.m.
14. (a); For last year; Lenovo = 100, Apple = 110
For this year; Lenovo = 120, Apple = 132
Samsung = 26.4

Last year, Samsung = 68.4

$$\text{Required percentage} = \frac{68.4}{278.4} \times 100 = 25\%$$

Hence, last year the sales of Samsung laptops were 25% of the total market for the laptops.

15. (b); Let solutions = 100 ml and therefore, alcohol = 40 ml.

$$\text{For first jar; } \frac{40+y}{100+y} = \frac{1}{2}$$

$$\therefore 80 + 2y = 100 + y \Rightarrow y = 100 - 80 = 20 \text{ ml}$$

$$\text{For second jar; } \frac{40 - \frac{2}{5}y + y}{60 - \frac{3}{5}y} = \frac{1}{1}, \quad y = \frac{100}{6}$$

Hence, required percentage

$$= \frac{20 - \frac{100}{6}}{\frac{100}{6}} \times 100 = 20\%$$

16. (c); Let the total number of students appeared for the test be y. Then,

$$\text{The number of male students appeared} = \frac{9y}{10}$$

$$\text{And the number of female students} = \frac{y}{10}$$

$$\text{Now, according to question} = \frac{3}{5} \times \frac{9y}{10} + \frac{4}{5} \times \frac{y}{10} = 1240$$

$$\frac{27y + 4y}{50} = 1240 \Rightarrow 31y = 1240 \times 50 = 62000$$

$$\therefore y = \frac{62000}{31} = 2000$$

Hence, the total number of candidates who appeared for the test is 2000.

17. (d); The best way of solving this question is through the method of elimination, i.e., going through the options.

Obviously, options (A), (B) and (C) cannot be the answer, because if 500 or more than 500 people are voting in against the resolution, then the motion cannot be passed.

Now, let us check the option (D). If 400 people are voting against the resolution for the first time, then 500 are voting in favour. Now, in the second voting, persons voting against the resolution would be 600, so persons voting in favour = 200.

In the first voting, resolution was passed by 100 votes and in the second voting, resolution was defeated by 400 votes. And it is 300% more than 100.

18. (a); Let the value of index = 100

Value of Vision Power share = 7

Value of Vision Infra = 13

Value of Vision Communication = 15

Value of remaining = 65

New value of Vision Power = 7.63

New value of Vision Infra = 14.3

New value of Vision Communication = 15.6

New value of remaining = 68.47

$$\text{Hence, required percentage} = \frac{68.47 - 65}{65} \times 100$$

$$= \frac{3.47}{65} \times 100 = 5.34\%$$

Hence, the price of other shares have increased by 5.34%

19. (b); Let x be A's profit and y be B's profit. Then,

$$x + 2520 = x \left(1 + \frac{10}{100}\right)^2 \Rightarrow x + 2520 = x \times \frac{121}{100}$$

$$100x + 252000 = 121x \Rightarrow 121x - 100x = 252000$$

$$\therefore x = \frac{252000}{21} = 12000$$

$$\text{Also, } 4200 = \frac{y \times 20 \times 1}{100} \Rightarrow 20y = 4200 \times 100$$

$$\therefore y = \frac{420000}{20} = 21000$$

C's profit = Rs. 9000

Ratio of their profit = 12000 : 21000 : 9000
= 12 : 21 : 9 = 4 : 7 : 3

$$\text{Therefore, C's share} = \frac{3}{14} \times 70000 = 3 \times 5000 = \text{Rs. 15000}$$

20. (d); Total money spent in buying goods = Rs. 63000

Total money kept in the bank = Rs. 27000

Total money received after selling the goods = Rs. 50400

Total money received from bank after 2 years = Rs. 35707.5

Therefore,

$$\text{Loss} = \text{Rs. } 90000 - \text{Rs. } (50400 + 35707.5) = \text{Rs. } 90000 - \text{Rs. } 86107.5 = \text{Rs. } 3892.5$$

$$\text{Hence, percentage loss} = \frac{3892.5}{90000} \times 100 = 4.325\%$$

Hence, the total assets of Sashi decreases by 4.325%

Previous Year (Memory Based)

1. (a); Let total marks = x

$$\therefore \frac{30x}{100} + 25 = \frac{40x}{100} - \frac{25}{100} \times \left(\frac{30x}{100} + 25 \right)$$

$$\frac{30x}{100} + 25 = \frac{40x}{100} - \frac{30x}{400} - \frac{25}{4}$$

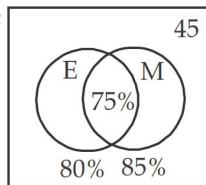
$$\frac{30x}{100} - \frac{40x}{100} + \frac{30x}{400} = \frac{-25}{4} - 25$$

$$\frac{120x - 160x + 30x}{400} = \frac{-25 - 100}{4}$$

$$\frac{-10x}{400} = \frac{-125}{4}, \quad x = \frac{40 \times 125}{4}, \quad x = 1250$$

$$\therefore \text{Maximum passing marks} = \frac{30}{100} \times 1250 + 25 = 400$$

2. (b);



Failed in both = 45

$$[100 - (80 + 85 - 75)]\% = 45$$

$$10\% \rightarrow 45$$

$$1\% \rightarrow \frac{45}{10} \Rightarrow 100\% \rightarrow 450$$

3. (b); $\frac{46}{100} \times 500 = \frac{32}{100} \times 300 + \frac{x}{100} \times 200$
 $230 = 96 + 2x, \quad 2x = 134, \quad x = 67\%$

4. (b); $[100 - (40 + 20 + 10 + 10)]\% \rightarrow 1500$

$$20\% \rightarrow 1500$$

$$1\% \rightarrow 75$$

$$100\% \rightarrow 7500$$

5. (d); $\left[100 - 20 - \frac{5}{100} \times 80 \right]\% \rightarrow 1400 + 120$

$$76\% \rightarrow 1520$$

$$1\% \rightarrow \frac{1520}{76}$$

$$4\% \rightarrow \frac{1520}{76} \times 4 \rightarrow 80 \text{ Rs.}$$

6. (d); Let total no of voters = 10,000

$$\text{Casting votes} = \frac{92}{100} \times 10,000 = 9200$$

$$\text{Winnig candidates votes} = 4800$$

$$\therefore 9200 - 4800 = 4400$$

$$\therefore 400 \rightarrow 1100$$

$$10,000 = \frac{1100}{400} \times 10,000 = 27500$$

7. (c); No. of inhabitants after 3 years

$$= 64000 \left(1 + \frac{5}{200} \right)^3 = 68921$$

8. (c); The value the machine 3 years

$$= 6250 \left(1 - \frac{10}{100} \right) \left(1 - \frac{20}{100} \right) \left(1 - \frac{30}{100} \right)$$

$$= 6250 \times \frac{9}{10} \times \frac{4}{5} \times \frac{7}{10} = 3150 \text{ Rs.}$$

9. (b); Sugar : Water

$$\begin{array}{ccc} 3 & : & 7 \Rightarrow \\ 7 & : & 3 \Rightarrow \end{array} \quad 40 \begin{array}{l} 9 : 21 \\ 49 : 21 \end{array}$$

$$\therefore 30 \rightarrow 75$$

$$40 \rightarrow \frac{75}{30} \times 40 = 100 \text{ gm}$$

10. (c); Let total of all items = $1200 + 1700 + 300 = 3200$

$$\text{After increment, total expense} = \frac{120}{100} \times 1200$$

$$+ \frac{130}{100} \times 1700 + \frac{150}{100} \times 300$$

$$= 1440 + 2210 + 450 = 4100$$

$$\text{Required} = \frac{4100 - 3200}{3200} \times 100$$

$$= \frac{9}{32} \times 100 = 28 \frac{1}{8}\%$$

11. (d); $62 \frac{1}{2}\%$ of 192 = 120

$$\text{Remaining hight} = 192 - 120 = 72$$

$$\therefore 12 \frac{1}{2}\% \text{ of } 72 = 9 \text{ m}$$

12. (d); A $\xrightarrow[3\text{ cm}]{5\text{ cm}} \underset{C}{\text{---}} \xrightarrow{2\text{ cm}} \text{B}$

$$\therefore \frac{106}{100} \times 3 = 3.18$$

$$\therefore \text{Required}\% = \frac{0.18}{2} \times 100 = 9\%$$

13. (a); A C D
100 80 100

$$\text{Required}\% = \frac{360}{500} \times 100 = 72\%$$

14. (b); Given ratio

$$\text{Women : Men} = 43 : 40$$

$$\text{Total person} = 3,11,250$$

$$\text{No. of women} = 43 \times 3750 = 1,61,250$$

$$\text{No. of Man} = 40 \times 3750 = 1,50,000$$

$$\text{No. of literate person}$$

$$= 161250 \times \frac{24}{100} + 150000 \times \frac{90}{100}$$

$$= 38700 + 135000 = 1,73,700$$

15. (b); Total population = 25000

$$\text{Female} = 5000, \quad \text{Male} = 20,000$$

$$\text{Uneducated persons} = 1000 + 2000 = 3000$$

$$\therefore \text{Required}\% = \frac{25000 - 3000}{25000} \times 100 = 88\%$$

16. (b); Total students = 140

$$\text{Passed students} = 48 + 48 = 96$$

$$\therefore \text{Passing}\% = \frac{96}{140} \times 100$$

$$= \frac{5}{7} \times 96 = \frac{480}{7} = 68\frac{4}{7}\%$$

17. (a); Required% = $\frac{12\frac{1}{2}}{112\frac{1}{2}} \times 100$

$$= \frac{25}{225} \times 100 = \frac{5 \times 100}{45} = \frac{100}{9} = 11\frac{1}{9}\%$$

18. (d); $30A = 25B = 20C$

$$A : B : C = 10 : 12 : 15$$

19. (a); $40(A + B) = 60(A - B)$

$$\frac{A+B}{A-B} = \frac{3}{2}$$

$$\text{Let } A = 2.5x, \quad B = 0.5x$$

$$\therefore \frac{2A - 3B}{A + B} = \frac{5 - 1.5}{3} = \frac{3.5}{3} = \frac{7}{6}$$

20. (a); $30 + 152 = x \times 637$
 $182 = 637x$

$$x = \frac{2}{7}$$

21. (d); $\frac{20}{100}A = \frac{30}{100}B = \frac{1}{6}C$

$$A : B : C = \frac{100}{20} : \frac{100}{30} : 6 = 5 : \frac{10}{3} : 6 = 15 : 10 : 18$$

22. (b); $\left(100 - 66\frac{2}{3}\right) = 33\frac{1}{3}\% = 1200$

$$\therefore 100\% = 3600$$

$$\therefore \text{Required expenses} = 66\frac{2}{3}\% \text{ of } 3600 \\ = 2400 \text{ Rs.}$$

23. (c);

Income	Expenditure	Saving
13500	9000	4500
15390	9630	5760

$$\% \text{ increase in his savings} = \frac{5760 - 4500}{4500} \times 100 \\ = 28\%$$

24. (d); X : Y : Z
300 : 150 : 100

$$\therefore \text{average} = \frac{300 + 150 + 100}{3} = \frac{550}{3}$$

$$\therefore \frac{550}{3} \rightarrow 110, \quad 300 \rightarrow \frac{110}{550} \times 3 \times 300 = 180 \text{ Rs.}$$

25. (a); Tulsi Ram Kashyap
120 : 100

$$\text{Tulsi Ram Salary} = \frac{720}{4} \times 100 = 18000$$

$$120 \rightarrow 18000, \quad 100 \rightarrow \frac{18000}{120} \times 100 = 15000 \text{ Rs.}$$

26. (a); No. = $96 \times 3 = 288$

$$67\% \text{ of the no} = \frac{67}{100} \times 288 = 192.96 \text{ Rs.}$$

27. (c); Let first part = x, second part = y

$$\frac{80x}{100} = \frac{60y}{100} + 3 \Rightarrow 4x - 3y = 15 \quad \dots (i)$$

$$\text{And, } \frac{80y}{100} = \frac{90x}{100} + 6 \Rightarrow 9x - 8y = -60 \quad \dots (ii)$$

$$\text{From (i) and (ii)}$$

$$x + y = 135$$

28. (d); Required % = $\frac{10}{90} \times 100 = \frac{100}{9} = 11\frac{1}{9}\%$

29. (d); Required % = $\frac{x}{100+x}\%$

30. (d); $\begin{array}{lcl} 100 & : & (100+r)\% \\ 100 & : & (100-r)\% \end{array}$

$\begin{array}{lcl} 1000 & : & 10000-r^2 \\ \therefore 10000-r^2 & \rightarrow & 1 \end{array}$

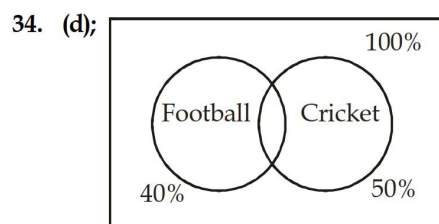
$\therefore 10,000 \rightarrow \frac{10,000}{10000-r^2}$

31. (c); Reduced price per kg of sugar = $\frac{83.7}{6.2} = 13.5$ Rs.

32. (a); Reduced Price = $\frac{64}{5} = 12.8$ Rs/kg

Original rate = $\frac{100}{80} \times 12.8 = \frac{128}{8} = 16$

33. (d); $100 \xrightarrow{-25\%} 75 \xrightarrow{+20\%} 90$
effect = -10%



Required % = $(40 + 50) - (100 - 18) = 8\%$

35. (b); Given mass of one = 8000 kg.

$\therefore \text{Metal} = \frac{60}{100} \times 8000 = 4800$ kg.

Silver = $\frac{3}{4}\%$ of metal = $\frac{3}{400}$ of metal

$\therefore \text{metal} = \frac{397}{400} \times 4800 = 4764$

36. (c); Let original fraction = $\frac{x}{y}$

$\therefore \frac{120x}{95y} = \frac{5}{2}$

$\frac{24x}{19y} = \frac{5}{2}, \quad \frac{x}{y} = \frac{19 \times 5}{24 \times 2} = \frac{95}{48}$

37. (c); $\frac{x \times 5 \times 1}{100} - \frac{(10000-x) \times 6 \times 1}{100} = 76.50$

$5x - 60,000 + 6x = 7650$

$11x = 67650 \Rightarrow x = 6150$ Rs.

$\therefore \text{Required amount} = (10,000 - 6150) = 3850$ Rs.

38. (c); Let his basic pay = x Rs.

$\therefore x + \frac{165x}{100} = 11925, \quad \frac{265x}{100} = 11925$

$x = 4500$ Rs.

39. (c); $\begin{array}{lcl} \text{Price} & 5 & : 6 \\ \text{Consumption} & 6 & : 5 \end{array}$

$\therefore \text{Required Ratio} = \frac{1}{6} = 1 : 6$

40. (b); $100 \xrightarrow{+50\%} 150$

$\therefore \text{Required Fraction} = \frac{50}{150} = \frac{1}{3}$